

Mokshith Sai Gadela

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SUMMARY

Software Engineer with experience developing scalable, highly available microservices and event-driven systems using **Kubernetes**, **AWS**, and **Kafka**. Skilled in building secure **REST APIs**, deploying **Machine Learning** models in **Python**, and optimizing relational databases to deliver high-performance enterprise applications.

EDUCATION

M.S. Computer Software Engineering, Arizona State University, Tempe, AZ

May 2026, GPA 3.9

- **Related Coursework:** Advanced Data Structure & Algorithms, Emerging Programming Paradigms, Foundations of Software Engineering, Data Mining, Statistical Machine learning, Mobile Systems and Software Project Process Management.

TECHNICAL SKILLS

- **Languages :** Python, Java, Go, C, C++
- **Web & Backend:** JavaScript, React, TypeScript, Node.js, Express.js, Spring Boot, Maven, Redis, Django
- **Cloud & Devops :** Kubernetes, Docker, AWS, Selenium, Jenkins, Prometheus, Terraform, GitHub Action, Kafka, OpenStack
- **Databases:** MySQL, PostgreSQL, Pandas
- **Tools:** Linux, Windows, Git, JIRA, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer, IJM Corporation, India

Aug 2022 - Jun 2024

- Built scalable backend **REST APIs** using **Node.js** and **TypeScript** to integrate supply chain modules and optimized complex **PostgreSQL** queries with **Redis** caching, cutting enterprise ERP report generation time from over 3 minutes to under 10 seconds.
- Migrated 5+ legacy monolithic services to a **Microservices** architecture on **Kubernetes (EKS)** utilizing custom **Docker** containers and **Helm** charts, establishing **Prometheus** for observability and reducing deployment cycles by 40%.
- Deployed supply chain prediction models using **Python** via **AWS Lambda** and **API Gateway** (Serverless), automating resource allocation logic and reducing manual scheduling discrepancies by an estimated 25%.
- Engineered an **event-driven** data pipeline leveraging **Apache Kafka** to process 10,000+ daily telemetry events, lowering ingestion latency to under 200ms for live asset tracking.
- Provisioned highly available cloud infrastructure (**VPC**, **RDS**, **S3**) using **Terraform** with strict **AWS IAM** Role-Based Access Control (RBAC), securing internal tools used by 50+ operators.

Software Engineer, Ditto Insurance, India

Feb 2022 - Jun 2022

- Automated **API** contract validation for the **JavaScript** frontend utilizing **Python**, **PyTest**, and **OpenAPI/Swagger** specifications to verify inbound **JSON** payloads, reducing staging environment crashes by 40%.
- Engineered **Python** data pipelines to extract and parse unstructured **JSON** telemetry from **Jenkins** and **Docker** logs, isolating infrastructure timeouts from defects across 5,000+ **CI/CD** weekly builds.

PROJECTS

Financial Education Platform | *Python, Django, React, AWS EC2, Jenkins*

- Delivered personalized financial insights with <2s latency by engineering a full-stack **React** and **Django** application, integrating **OpenAI APIs** to process and dynamically render user spending data.
- Accelerated API response efficiency by 1.5x and achieved zero-downtime updates by optimizing **REST API** data serialization with **JWT** session management, and orchestrating a **Jenkins CI/CD** deployment pipeline to **AWS EC2**.

Custom Scheduling Operator | *Go, Kubernetes, Docker*

- Engineered a production-grade **Kubernetes** operator in **Go** using **CRDs** and admission webhooks, reconciling 5,000 pods and reducing compute costs by 30% through intelligent scheduling logic.
- Orchestrated full build and deployment pipeline with **Docker** and **Make** demonstrating end-to-end ownership of a platform-level infrastructure component.

Decentralized Key-Value Store | *Go, Docker, Grafana*

- Processed 15,000+ ops/sec across a **Docker**-containerized cluster by engineering a **Go**-based gossip protocol over **gRPC** for decentralized failure detection.
- Restored consistency in <100ms during network partitions simulated via **Chaos Mesh**, utilizing **Prometheus** and **Grafana** to track latency metrics while CRDTs resolved concurrent cluster writes.

Learning-Based Cache Optimization | *Python, PyTorch, GRU*

- Trained a Gated Recurrent Unit (**GRU**) sequence model using **PyTorch** to predict memory access patterns across trace datasets containing over 1 million requests.
- Engineered feature extraction pipelines using **Pandas** to isolate temporal locality metrics, increasing the cache hit rate by 12% compared to standard LRU baselines.

CERTIFICATIONS

- **AWS Certified Solutions Architect - Associate (SAA-C03)**
- **Kubernetes** Fundamentals Certification - The Linux Foundation
- Elite Certification in Programming, **Data Structures and Algorithms** - NPTEL IIT Madras

ACTIVITIES

Lead Web Developer, EDAM Technical Club, India

April 2021 - Jan 2022

- Taught web development fundamentals to 50+ freshman students through live coding sessions focusing on **JavaScript**, DOM manipulation, and consuming **REST APIs**.